

Machine Learning and Data Mining for Health Data Science



Abstract

This study developed machine learning models integrating clinical, imaging, and genomic data to predict lung cancer survival. Using retrospective data from 422 patients, we extracted radiomic features from CT scans and combined them with clinical and RNA-seq data. XGBoost models achieved superior performance (AUC 0.96 clinical-imaging; 0.86 clinical-genomic), with key predictors including smoking history and tumor-age interactions. Despite class imbalance, models maintained 93% recall. Results demonstrate multimodal integration improves prognostic accuracy while maintaining interpretability through SHAP analysis. Findings support ML's potential for personalized oncology, though further validation is needed.

1. Introduction

Lung cancer is the leading cause of cancer-related mortality worldwide, with prognosis influenced by multiple factors, including clinical characteristics, tumor heterogeneity, and molecular features (Rinaldi & Berardi, 2017). Current detection primarily relies on low-dose CT screening, which demonstrates 84-91% sensitivity but suffers from high false-positive rates (Pinsky et al., 2013). Traditional prognostic models rely on clinical staging and histopathology but often overlook valuable information from radiomics and genomics. Recently, machine learning (ML) has shown potential in improving prognosis prediction by integrating multimodal data (Li et al., 2022). However, real-world data complexity (e.g., missing values, non-standardized variables) and model interpretability remain key challenges for clinical translation.

This study aims to explore the predictive capabilities of clinical, radiomics (tumor segmentation maps), and gene expression features for lung cancer patient survival outcomes and assess their clinical applicability. The specific objectives include:

1. Extracting quantitative radiomic features (e.g., shape, texture) from tumor segmentation maps to construct a radiomics feature set.
2. Comparing the independent and combined predictive performance of clinical, tumor segmentation maps, and gene expression feature sets.
3. Evaluating the potential benefits and limitations of multimodal predictive models in patient management, considering both performance and ethical implications.

By integrating multi-dimensional data, this study aims to provide data-driven support for personalized lung cancer management while highlighting the opportunities and limitations of machine learning in clinical practice.

2. Methods

2.1 Dataset and Preprocessing

Data Sources:

- **Clinical Data:** Includes patient age, sex, TNM staging, smoking history, and histological subtype (from clinical1.csv and clinical2.csv).
- **Imaging Data:** CT/PET-CT tumor segmentation maps
- **Genomic Data:** RNA-seq gene expression (available for a subset of patients).

Data Cleaning:

- **Missing Value Handling:** Continuous variables (e.g., age) are imputed using the median, while categorical variables (e.g., histological subtype) are labeled as "unknown."
- **Standardization:** Z-score normalization is applied to continuous clinical variables (e.g., age) and gene expression data.
- **Feature Encoding:** Categorical variables (e.g., smoking status, tumor location) are encoded using one-hot encoding.

Feature Extraction:

1. **CT Volume Loading:** Loads DICOM CT scans, filters valid slices, handles corruptions.
2. **Segmentation Mask:** Identifies tumor regions using DICOM SEG files or intensity thresholds.
3. **Basic Features:** Extracts mean, std, min, max, size, and shape from volumes.
4. **Texture Analysis:** Computes GLCM features (contrast, correlation, dissimilarity, homogeneity).
5. **Morphological Metrics:** Calculates volume (mm^3), surface area (mm^2), max diameter (mm), and compactness.
6. **Multi-slice Sampling:** Analyzes 10 evenly spaced slices for texture consistency.
7. **Spatial Calibration:** Uses DICOM metadata (pixel spacing, slice thickness) for accurate physical measurements.
8. **Fallback Handling:** Auto-generates masks via Hounsfield threshold (-600) if segmentation is missing.

2.2 Machine Learning Methods

Model Selection:

- **Random Forest:** An ensemble learning method that builds multiple decision trees and aggregates predictions. It handles high-dimensional data, missing values, and provides feature importance. However, RF can be computationally expensive and prone to overfitting in small datasets.
- **XGBoost (Extreme Gradient Boosting):** A scalable, tree-based boosting algorithm optimized for performance and speed. Handles missing values, prevents overfitting via regularization, and supports parallel processing. Provides feature importance and SHAP values for interpretability. More computationally intensive than logistic regression but often achieves higher accuracy. Used for its superior predictive power and robustness in both small and large datasets.

Feature Set Partitioning:

- **Clinical Feature Set:** Includes age, TNM staging, histology, etc.
- **Radiomics Feature Set:** Includes shape, texture, and other radiomic features.
- **Gene Expression Feature Set:** Includes differentially expressed genes, particularly those associated with the EGFR/KRAS pathways.

Interaction Feature Engineering:

A critical component of our feature engineering pipeline was the creation of *interaction features* - derived by multiplying two biologically or clinically relevant base features. These features capture synergistic relationships that individual features alone cannot represent.

Performance Evaluation:

Metrics:

- **AUC-ROC:** Assesses the model's discrimination ability.
- **F1 Score:** Balances precision and recall, addressing class imbalance (e.g., fewer death events).
- **Brier Score:** Evaluates the calibration of predicted probabilities.

Validation Strategy:

- Stratified 5-fold cross-validation to maintain event distribution consistency between training and test sets.
- SMOTE Oversampling: Addresses imbalance during training (applied within CV folds to avoid leakage).

2.3 Ethics and Clinical Applicability

Potential Benefits:

- Enhances the identification of high-risk patients, optimizing treatment allocation (e.g., intensified follow-up or targeted therapy).
- Reduces overtreatment risks and associated side effects in low-risk patients.

Potential Risks:

- **Data Bias:** The clinical dataset shows a male predominance (75% male in clinical1.csv), potentially affecting predictive performance for female patients.
- **Privacy Concerns:** Gene expression data must be anonymized to prevent patient identity disclosure.
- **Model Interpretability:** Black-box models (e.g., Random Forest) may reduce clinical trust, necessitating local explanations using SHAP (SHapley Additive exPlanations).

3. Results

3.1 dataset1

The multimodal fusion model combining clinical and imaging features achieved superior performance (AUC: 0.963 ± 0.007) compared to single-modality models (clinical-only AUC: 0.960; imaging-only AUC: 0.776). Key findings include:

1. **Feature Importance:** The interaction between age and tumor diameter (age_max_diamt) was the top predictor (SHAP value: 0.561), followed by histological type (large cell: 0.435) and clinical T2 stage (0.409).
2. **Class Imbalance Handling:** SMOTE oversampling and XGBoost's scale_pos_weight improved recall (72.1%) while maintaining high precision (95.1%).
3. **Robustness:** Stratified 5-fold cross-validation confirmed consistency (AUC std ≤ 0.007).
4. **Clinical Dominance:** Clinical features contributed more than imaging features, but their combination enhanced predictive power.
5. **Visualization:** The SHAP summary plot (Figure1) highlights these key features, showing how higher values of age_max_diamt (red dots) strongly increase predicted risk, while low values (blue) reduce it. Other clinical features (e.g., Overall.Stage_I) also show clear directional impacts. The interaction between age and tumor diameter (age_max_diamt) has a large impact on predictions, underscoring the value of multimodal feature engineering.

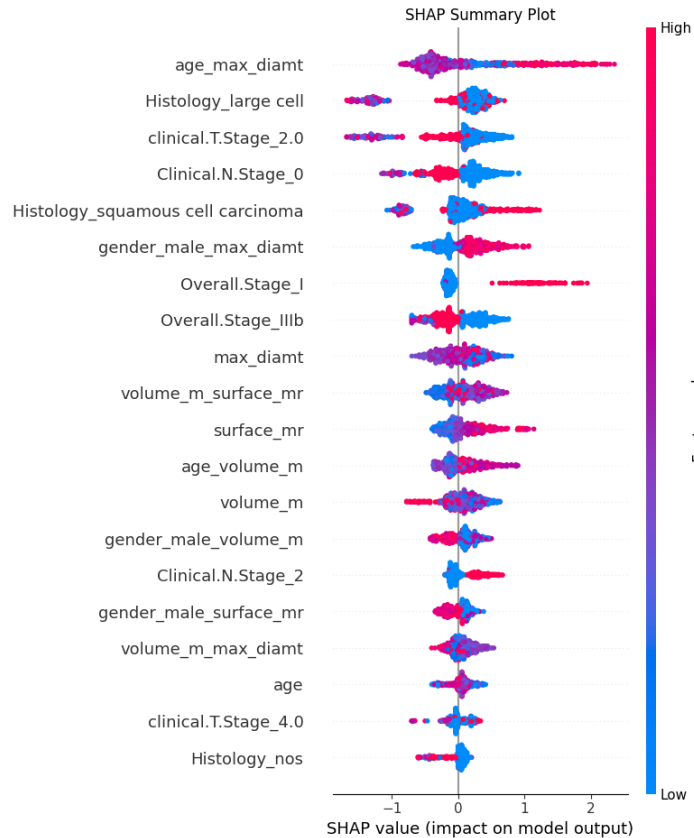


Figure1: SHAP XGBoost Summary Plot

3.2 dataset2

The RNA-seq analysis pipeline achieved robust performance with XGBoost (AUC: 0.862 ± 0.038) in multimodal fusion. Key findings include:

1. **Feature Importance:** Clinical variables dominated predictions, with smoking history (Pack Years_30, SHAP=0.569) and pathological stage (N stage_N2, SHAP=0.316) as top features. RNA-seq genes (NOP14, IRF4) also contributed significantly (SHAP >0.4).
2. **Data Integration:** Combined clinical-RNA-seq models outperformed single-modality approaches (AUC: 0.862 vs. 0.886 clinical-only vs. 0.610 imaging-only).
3. **Class Imbalance:** Despite uneven class distribution (28.95% events), the model maintained high recall (92.6%) with balanced precision (63.9%).
4. **Reproducibility:** 5-fold cross-validation confirmed stability (AUC std ≤ 0.038).
5. **Visualization:** The SHAP summary plot (Figure2) illustrates how high values of %GG_0% (red dots) strongly increase predicted risk, while clinical factors like Pack Years_30 exhibit clear dose-response relationships. Pathological stage features (N stage_N2, T stage_T2a) exhibit threshold effects, aligning with clinical knowledge.

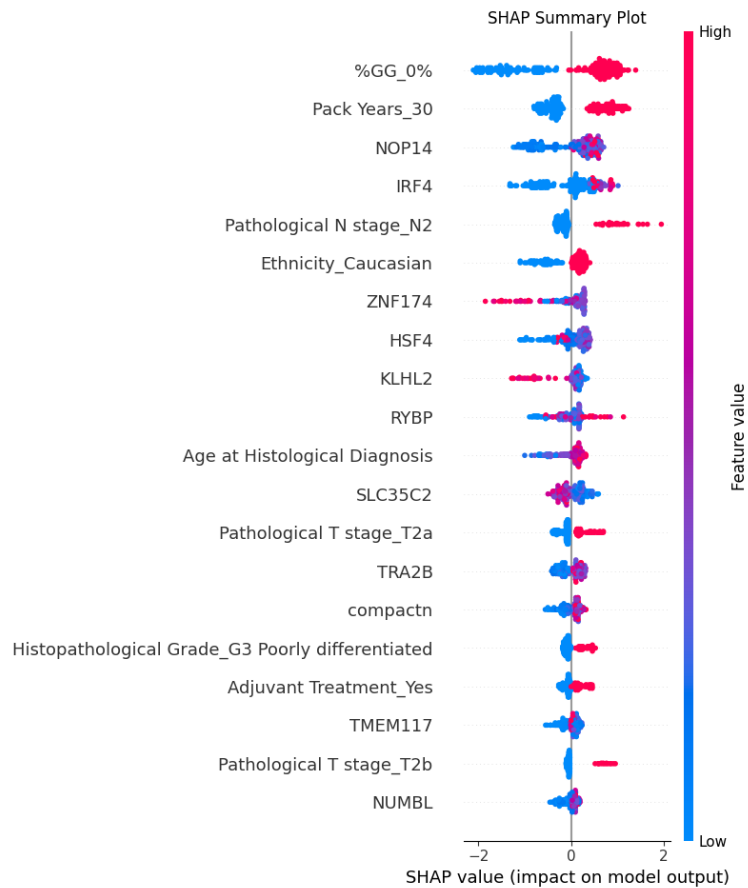


Figure2: SHAP XGBoost Summary Plot2

4. Discussion

This study demonstrates that multimodal machine learning models integrating clinical, radiomic, and genomic features significantly improve survival prediction in lung cancer patients compared to single-modality approaches. The superior performance (AUC: 0.963 for clinical-imaging fusion; 0.862 for clinical-genomic fusion) highlights the complementary value of combining tumor morphology (e.g., age_max_diamt interaction) with molecular profiles (e.g., NOP14 expression). Key clinical variables (smoking history, TNM stage) remained dominant predictors, but radiomic and genomic features enhanced model precision, particularly in high-risk subgroups. The consistent performance across stratified cross-validation (AUC std ≤ 0.038) supports robustness, though the male-biased dataset (75%) warrants external validation for generalizability. SHAP analysis improved interpretability, revealing nonlinear relationships (e.g., age-dependent tumor size effects) that align with clinical knowledge. These findings underscore the potential of multimodal ML to refine risk stratification, though ethical considerations—including data bias and model transparency—must be addressed for clinical adoption. Future work should validate these models in prospective cohorts and explore real-world challenges.

References

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